

Lessons

Multiplying and Dividing Numbers with Signs

Multiplication is fast addition. In other words, you can multiply when you have to add the same number many times.

5 + 5 is the same as 2 × 5. Both equal 10.

(−5) + (−5) is the same as 2 × (−5). Both equal −10.

Multiplying a negative number by a negative number is hard to visualize. You can apply your reasoning skills to find out how it works.

What number works in the blank? −5 × −2 = 10

or $(-5) \cdot (-2) = 10$
or $(-5)(-2) = 10$

If you put a positive number in the blank, the product would have to be −10. A positive number won’t work. The missing number must be negative: −5 × −2 = 10

Remember that multiplication can be written in three ways: using a times symbol (×), using a raised dot (•), or writing the numbers next to each other in parentheses.

In algebra, division is written using a fraction bar. For example, 15 ÷ 5 is written $\frac{15}{5}$.

example: $-10 \div (-2) = 5$ or $\frac{-10}{-2} = 5$

Tip

***** Use these rules to multiply or divide two signed numbers:

If the signs are the same, the answer is positive. 3 × 4 = 12 −3 × −4 = 12

$$\frac{3}{4} = 0.75$$

$$\frac{-3}{-4} = 0.75$$

If the signs are different, the answer is negative. −3 × 4 = −12 3 × −4 = −12

$$\frac{3}{-4} = -0.75$$

$$\frac{-3}{4} = -0.75$$

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